HOPF BIFURCATION

CARLANGELO LIVERANI

For $\lambda \in [-1, 1]$ let us consider the family of differential equations

$$\dot{x} = \begin{pmatrix} \lambda & \omega \\ -\omega & \lambda \end{pmatrix} x + V(x) =: A_\lambda x + V(x) =: V_\lambda(x) \tag{0.1}$$

where $x \in \mathbb{R}^2$, $\omega > 0$ and $V \in \mathcal{C}^3(\mathbb{R}^n, \mathbb{R}^n)$, $\|\nabla V\|_\infty < \infty$, so the equation admits global solutions in time, that is it defines a flow. Suppose further $V(0) = 0$, $\nabla_0 V = 0$, and $D_0^2 V \neq 0$.

Clearly the case $\lambda = 0$ it is not a generic vector field, yet if we consider the family V_λ of vector fields then it is easy to very that each small perturbation will have an element with linear part with zero real part, thus, as a family, we have a generic situation.

Condition 1. *We assume that, for $\lambda < 0$, the only invariant set in a neighborhood of the origin is an attracting focus.*

Exercize 1. *Show that*

$$e^{A_\lambda t} = e^{\lambda t} \begin{pmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{pmatrix} =: e^{\lambda t} R(\omega t).$$

Exercize 2. *Show that the solutions $x(t, \lambda, z)$ of (0.1) with $x(0, \lambda, z) = (0, z)$ satisfy*

$$x(t, \lambda, z) = e^{\lambda t} R(\omega t)(0, z) + \int_0^t e^{\lambda(t-s)} R(\omega(t-s)) V(x(s, \lambda, z)) ds \tag{0.2}$$

Moreover, show that there exists $\lambda_0, \delta, C > 0$ such that $\|x(t, \lambda, z)\| \leq C|z|$ for all $|t| \leq \frac{2\pi}{\omega}$, $|\lambda| \leq \lambda_0$ and $|z| \leq \delta$. Finally, show that $x \in \mathcal{C}^2(\mathbb{R}^3)$. (Hint: work out the proof of existence via the usual fixed point argument. Then use the smooth dependence form the parameters and the initial conditions.)

The idea is to study, when varying λ, the Poincarè map induced by the flow, in a neighborhood of zero, on the vertical section $S := \{x \in \mathbb{R}^2 \;:\; \|x\| \leq \delta;\; \langle x, e_1 \rangle = 0\}$. Clearly, for $|(0, z)| \leq \delta/2$, and λ small enough there will exists times $\bar{\tau}_i(\lambda, z)$ such that the trajectory visits again S. Such a times will satisfy the equation

$$F_0(t, \lambda, z) := \langle x(t, \lambda, z), e_1 \rangle = 0.$$

Since for V and λ equal zero the trajectories describe circles, when it intersect S the first time it has described an half circle, it then natural to look at the second intersection with s, the time $\tau(\lambda, z) = \bar{\tau}_2(\lambda, z)$. Clearly, if $V = 0$ then $\tau(\lambda, z) = \frac{2\pi}{\omega}$.

The idea is to determine τ via the implicit function theorem, for small z, but since $F_0(\frac{2\pi}{\omega}, \lambda, 0) = 0$ the function is degenerate for $z = 0$. To overcome this

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problem it is helpful to define $\xi(t,\lambda,z) = z^{-1}x(t,\lambda,z)$. Clearly,

$$\xi(t,\lambda,z) = e^{\lambda t}R(\omega t)e_2 + \int_0^t e^{\lambda(t-s)}R(\omega(t-s))z^{-1}V(z\xi(s,\lambda,z))ds. \tag{0.3}$$

Exercize 3. *Show that, for all* $|t| \leq \frac{3\pi}{\omega}$, $|\lambda| \leq \lambda_0$ *and* $|z| \leq \delta$,

$$\xi(t,\lambda,z) = e^{\lambda t}R(\omega t)e_2 + \alpha(t,\lambda)z + z^2\beta(t,\lambda,z)$$

with $\alpha \in \mathcal{C}^\infty$ *and* $\beta \in \mathcal{C}^2$. *In addition shows that* $\alpha(\frac{2\pi}{\omega},0) = 0$ *and, generically,* $\langle\beta(\frac{2\pi}{\omega},0,0),e_2\rangle \neq 0$.

(Hint: By hypotheses and Exercise 2

$$z^{-1}V(z\xi(s,\lambda,z)) = z\langle\xi(s,\lambda,z),B\xi(s,\lambda,z)\rangle + \mathcal{O}(z^2).$$

and $\|\xi(s,\lambda,z)\| \leq C$. *Next, notice that*

$$\begin{aligned}\alpha\left(\frac{2\pi}{\omega},0\right) &= \int_0^{\frac{2\pi}{\omega}} R(\omega s)D_0V(R(\omega s)e_2,R(\omega s)e_2)ds\\ &= \omega^{-1}\int_0^{2\pi} R(s)D_0V(R(s)e_2,R(s)e_2)ds.\end{aligned}$$

The above integral boils down to the integrals of products of three trigonometric functions that are all zero. To see the last fact is suffices to notice that if the quantity is zero a small change in D_0^2V *will make it different from zero.)*

Condition 2. *We assume* $A := \langle\beta(\frac{2\pi}{\omega},0,0),e_2\rangle \neq 0$.

We can then define $F_1(t,\lambda,z) := z^{-1}F_0(t,\lambda,z) = \langle\xi(t,\lambda,z),e_1\rangle$. Clearly for $z \neq 0$ the equations $F_0 = 0$ and $F_1 = 0$ are equivalent. Yet, we have $F_0(t,\lambda,0) = 0$, while $F_1(t,\lambda,0) = e^{\lambda t}\langle R(\omega t)e_2,e_1\rangle$, so $F_1(\frac{2\pi}{\omega},\lambda,0) = 0$.

To apply the implicit function theorem we need to check that $\partial_tF_1(\frac{2\pi}{\omega},\lambda,0)$ is uniformly bounded away from zero.

Exercize 4. *Show that*

$$\partial_tF_1(\frac{2\pi}{\omega},\lambda,0) = \langle\dot{\xi}(\frac{2\pi}{\omega},\lambda,0),e_1\rangle = -\omega e^{\lambda\frac{2\pi}{\omega}}.$$

So the implicit function theorem can be applied yielding a $\mathcal{C}^2$ solution $\tau(\lambda,z)$.

Exercize 5. *Show that*

$$\tau(\lambda,z) = \frac{2\pi}{\omega} + \lambda a(\lambda)z + b(\lambda,z)z^2,$$

where $\frac{\langle\alpha(\frac{2\pi}{\omega},\lambda),e_1\rangle}{\omega\lambda e^{2\frac{\lambda\pi}{\omega}}}$, *provided we choose* δ,λ_0 *small enough.*

We can then define the Poincarè maps $T_\lambda : S \to S$ defined by

$$\begin{aligned}T_\lambda(z) &:= \langle x(\tau(\lambda,z),\lambda,z),e_2\rangle\\ &= ze^{\lambda\tau(\lambda,z)}\cos(\lambda\omega a(\lambda)z) + \langle\alpha(\tau(\lambda,z),e_2\rangle z^2 + \langle(\beta(\tau(\lambda,z),\lambda),e_2\rangle z^3 + \mathcal{O}(z^4).\end{aligned}$$

Exercize 6. *Show that there exist a* $\mathcal{C}^1$ *function* $c : \mathbb{R} \to \mathbb{R}$ *such that*

$$T_\lambda(z) = e^{\frac{2\pi\lambda}{\omega}}z + \lambda c(\lambda)z^2 + Az^3 + \mathcal{O}(\lambda z^3 + z^4).$$

In addition, show that the solutions, in a neighborhood of zero, of $T_\lambda(z) = z$ *are* $z = 0$ *and* $z = \pm\sqrt{\frac{2\pi\lambda}{\omega A}} + \mathcal{O}(\lambda)$.

Since for $\lambda < 0$ the original equation has only an attracting focus at the origin and no periodic orbit can be present it follows that it must be $A > 0$. Accordingly, for $\lambda > 0$ we have two fixed point that must correspond to a periodic orbit for the original system. Finally, such a periodic orbit is attracting, indeed

Exercize 7. *Prove that* $|DT_\lambda| < 1$ *at the fixed points.*

Remark 0.1. *The above scenario is called* Hopf bifurcation *and happens generically each time that the real part of a fixed point (with a non zero imaginary part) crosses zero: an attracting focus becomes a repelling focus but it is surrounded by an attracting periodic orbit.*

1. A (slightly) different point of view

Given the fact that the trajectories rotate around zero almost in circles it may occur the idea to treat the problem in polar coordinates. In fact this point of view is quite advantageous and we will carry it out in order to show how a problem may simplify if viewed in different coordinates.

The polar coordinates can be written as $x = \rho v(\theta)$, where $\rho \in \mathbb{R}_+$, $\theta \in [0, 2\pi]$ and $v(\theta) := (\cos\theta, \sin\theta)$.

Remark 1.1. *Note that such a change of coordinates is singular for* $\rho = 0$

If we substitute such coordinates in (0.1), we obtain

$$\dot\rho v(\theta) + \rho n(\theta)\dot\theta = \lambda\rho v(\theta) - \omega\rho n(\theta) + V(\rho v(\theta)),$$

where $n(\theta) := (-\sin\theta, \cos\theta)$. That is

$$\begin{aligned} \dot\rho &= \lambda\rho + \langle v(\theta), V(\rho v(\theta))\rangle =: \lambda\rho + a(\theta,\rho)\rho^2 \\ \dot\theta &= -\omega + \rho^{-1}\langle n(\theta), V(\rho v(\theta))\rangle =: -\omega + b(\theta,\rho)\rho. \end{aligned} \tag{1.1}$$

Note that the equation (1.1) is well defined also for $\rho = 0$ but in such a case, instead of a fixed point, it has the periodic orbit $(\rho(t), \theta(t)) = (0, -\omega t)$. In some sense the polar coordinates have automatically regularized the behaviour at zero saving us the trouble to do it by hand as we did in the Cartesian coordinates. Since for small ρ we have $\dot\theta < 0$, it is convenient to use θ rather than t to parameterize the motion. Calling again ρ the distance from the origin as a function of θ we have

$$\frac{d\rho}{d\theta} = \frac{\lambda\rho + a(\theta,\rho)\rho^2}{-\omega + b(\theta,\rho)\rho} =: -\frac{\lambda}{\omega}\rho - \beta(\theta,\lambda)\rho^2 - \gamma(\theta,\rho,\lambda)\rho^3, \tag{1.2}$$

where

$$\begin{aligned} &\beta(\theta,\lambda) = \omega^{-1}a(\theta,0) + \lambda\omega^{-2}b(\theta,0) \\ &\gamma(\theta,0,\lambda) = \lambda b(\theta,0)^2\omega^{-3} + a(\theta,0)b(\theta,0)\omega^{-2} + \lambda\partial_\rho b(\theta,0)\omega^{-1} + \partial_\rho a(\theta,0)\omega^{-1}. \end{aligned}$$

At this point the Poincarè section corresponds simply at looking at the trajectory for multiple of 2π. We wish then to define the map $S_\lambda : \mathbb{R}_+ \to \mathbb{R}_+$ defined by $S_\lambda\xi := \rho(2\pi, \xi, \lambda)$, where $\rho(2\pi, \xi, \lambda)$ is the solution of (1.2) with initial condition ξ seen at $\theta = 2\pi$.[1]

[1] Since $\dot\theta < 0$, here we are going back in time with respect to the previous section, hence S_λ is essentially what before was T_λ^{-1}.

Remark that (1.2) can be written in integral form as

(1.3)
$$\rho(\theta,\xi,\lambda)=e^{-\frac{\lambda}{\omega}\theta}\xi-\int_0^\theta e^{-\frac{\lambda}{\omega}(\theta-\varphi)}\left[\beta(\varphi,\lambda)\rho(\varphi,\xi,\lambda)^2-\gamma(\varphi,\rho(\varphi,\xi,\lambda),\lambda)\rho(\varphi,\xi,\lambda)^3\right]d\varphi$$

Iterating (1.3) yields

(1.4)
$$\begin{aligned}\rho(\theta,\xi,\lambda)=&e^{-\frac{\lambda\theta}{\omega}}\xi-\int_0^\theta e^{-\frac{\lambda(\theta-\varphi)}{\omega}}\beta(\varphi,\lambda)\left[e^{-\frac{2\lambda\varphi}{\omega}}\xi^2-2\int_0^\varphi e^{-\frac{\lambda(2\varphi-3x)}{\omega}}\beta(x,\lambda)\xi^3\right]d\varphi\\&-\int_0^\theta e^{-\frac{\lambda(\theta-\varphi)}{\omega}}\gamma(\varphi,0,\lambda)e^{-\frac{3\lambda\varphi}{\omega}}\xi^3d\varphi+\mathcal{O}(\xi^4).\end{aligned}$$

To compute the map S_λ we must set $\theta=2\pi$ in (1.4). As already noted in the previous section some integral will be zero. Indeed, from (1.1) follows that $a(\theta,0),b(\theta,0)$ are homogeneous polynomials of third order while $\partial_\rho a(\theta,0),\partial_\rho b(\theta,0)$ are of forth order. Moreover, setting

$$\Gamma(\theta,\lambda):=\int_0^\theta\beta(\varphi,\lambda)d\varphi,$$

we have $\Gamma(2\pi)=0$, hence

$$\int_0^{2\pi}\beta(\varphi,\lambda)\int_0^\varphi\beta(x,\lambda)dxd\varphi=\int_0^{2\pi}\Gamma'(\varphi,\lambda)\Gamma(\varphi,\lambda)d\varphi=\frac{\Gamma(2\pi,\lambda)^2-\Gamma(0,\lambda)^2}{2}=0.$$

Given the above considerations we can finally write

$$S_\lambda(\rho)=e^{-\frac{2\pi\lambda}{\omega}}\rho+B\rho^3+\mathcal{O}(\lambda\rho^2+\rho^4). \tag{1.5}$$

where

$$B=-\int_0^{2\pi}\left[a(\theta,0)b(\theta,0)\omega^{-2}+\partial_\rho a(\theta,0)\omega^{-1}\right]d\theta$$

Exercize 8. *Compute, in terms of the Tailor coefficients of V, what it means $B\neq 0$.*

As before the fixed points of S_λ are $\rho=0$ and $\rho_p:=\sqrt{\frac{2\pi\lambda}{\omega B}}+\mathcal{O}(\lambda)$ which exists only if $\lambda B>0$. In the latter case $S_\lambda'(\rho_p)=1+\frac{4\pi\lambda}{\omega}+\mathcal{O}(\lambda^{\frac{3}{2}})$, while $S_\lambda'(0)=1-\frac{2\pi\lambda}{\omega}+\mathcal{O}(\lambda^2)$. Thus, if one is attracting the other must be repelling. If, for $\lambda B<0$, we want only an attracting focus (as we asked in the previous section), then it must be $B<0$. Hence, for $\lambda>0$ we have an attracting focus, while for $\lambda>0$ the focus becomes repelling and it appears an attracting periodic orbit.

Carlangelo Liverani, Dipartimento di Matematica, II Università di Roma (Tor Vergata), Via della Ricerca Scientifica, 00133 Roma, Italy.

E-mail address: liverani@mat.uniroma2.it